

Healthcare Analytics for Improving Patient Outcomes

1. Project Overview

This project focuses on analysing employee performance and retention patterns within an organisation. By leveraging HR data, the goal is to identify key factors influencing employee productivity, performance, and retention, and provide actionable insights for improving workforce efficiency, reducing turnover, and fostering a positive work environment. The project will employ data analysis techniques, including exploratory data analysis (EDA), trend analysis, and visualisation to help HR managers make data-driven decisions for workforce management.

2. Dataset Summary

- Rows: 55500
- Columns: 15
- Key Features :
 - Patient Demographics (Name, Age Gender, Blood Type)
 - Clinical Information (Medical Condition, Medication, Test Results)
 - Hospital & Administrative Details (Doctor, Hospital, Room Number)
 - Admission & Discharge Details(Date of Admission, Discharge Date, Admission Type)
 - Financial Information (Insurance Provider, Billing Amount)
- Missing Data: The dataset does not have any missing data

3. Workforce Overview

Employees are mostly in the early to mid-career range (20–40 years).

Distribution across departments and regions varies, highlighting operational concentration risks.

Recruitment channels influence performance outcomes, indicating differences in candidate quality.

4. Performance Insights

High training scores, KPI achievement, and awards strongly correlate with better performance.

Consistent performers continue to excel year-over-year.

Departments with lower performance may need targeted interventions.

5. Retention Trends

High retention: Employees aged 30–40, award winners, and consistent KPI achievers.

Retention risk: Early-career employees (20–30), low KPI achievers, and employees in certain departments with shorter tenure.

Effective training and recognition directly improve retention.

6. Exploratory Data Analysis (EDA)

We began with data preparation and cleaning in Python:

- Data Loading: Imported the dataset using pandas.
- Initial Exploration: Used `df.info()` to check structure and `.describe()` for summary statistics.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 55500 entries, 0 to 55499
Data columns (total 15 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Name                                  55500 non-null  object
1   Age                                   55500 non-null  int64
2   Gender                               55500 non-null  object
3   Blood Type                           55500 non-null  object
4   Medical Condition                    55500 non-null  object
5   Date of Admission                    55500 non-null  object
6   Doctor                               55500 non-null  object
7   Hospital                             55500 non-null  object
8   Insurance Provider                   55500 non-null  object
9   Billing Amount                        55500 non-null  float64
10  Room Number                          55500 non-null  int64
11  Admission Type                       55500 non-null  object
12  Discharge Date                       55500 non-null  object
13  Medication                           55500 non-null  object
14  Test Results                         55500 non-null  object
dtypes: float64(1), int64(2), object(12)
memory usage: 6.4+ MB

```

	Age	Billing Amount	Room Number
count	55500.000000	55500.000000	55500.000000
mean	51.539459	25539.316097	301.134829
std	19.602454	14211.454431	115.243069
min	13.000000	-2008.492140	101.000000
25%	35.000000	13241.224655	202.000000
50%	52.000000	25538.069380	302.000000
75%	68.000000	37820.508432	401.000000
max	89.000000	52764.276740	500.000000

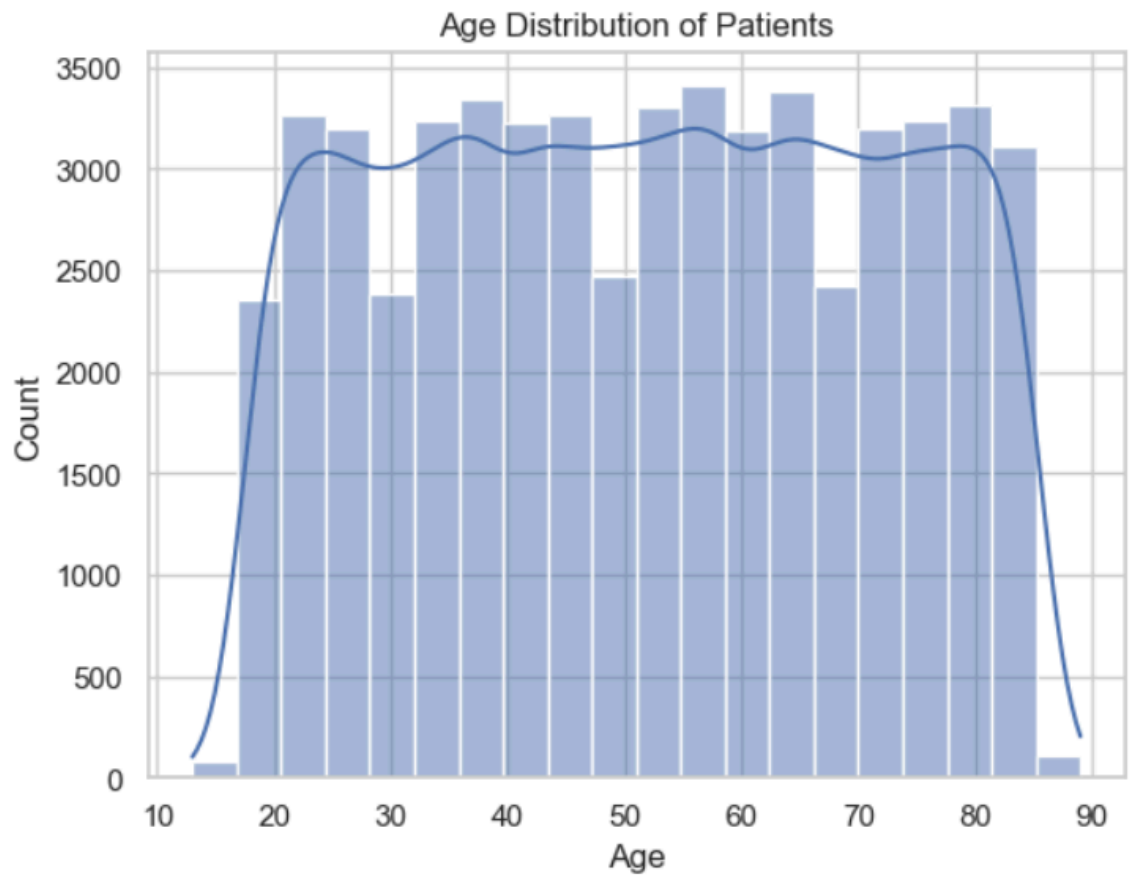
```
healthcare.describe(include = 'all')
```

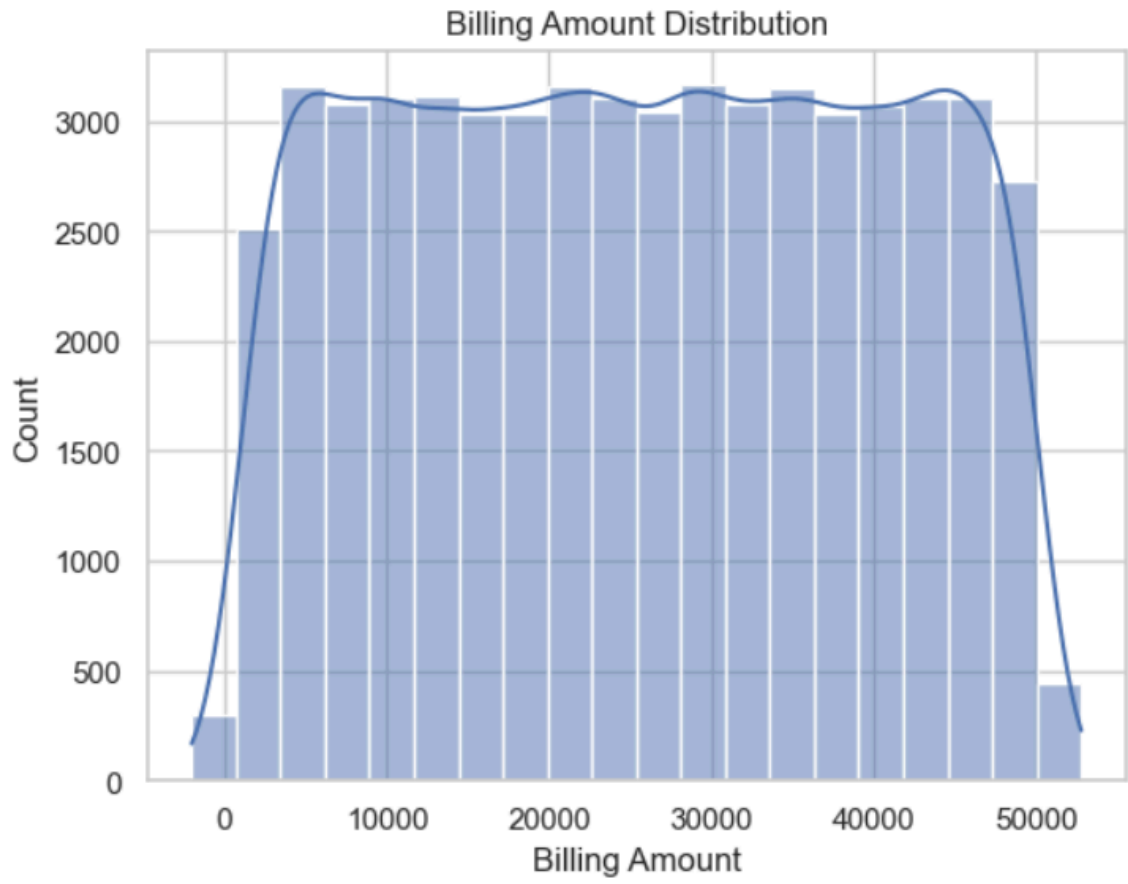
	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor	Hospital	Insurance Provider	Billing Amount	Room Number	Admission Type	Discharge Date
count	55500	55500.000000	55500	55500	55500	55500	55500	55500	55500	55500.000000	55500.000000	55500	55500
unique	49992	NaN	2	8	6	NaN	40341	39876	5	NaN	NaN	3	NaN
top	DAvid muNoZ	NaN	Male	A-	Arthritis	NaN	Michael Smith	LLC Smith	Cigna	NaN	NaN	Elective	NaN
freq	3	NaN	27774	6969	9308	NaN	27	44	11249	NaN	NaN	18655	NaN
mean	NaN	51.539459	NaN	NaN	NaN	2021-11-01 01:02:22.443243008	NaN	NaN	NaN	25539.316097	301.134829	NaN	2021-11-1 13:15:20.82162150
min	NaN	13.000000	NaN	NaN	NaN	2019-05-08 00:00:00	NaN	NaN	NaN	-2008.492140	101.000000	NaN	2019-05-0 00:00:00
25%	NaN	35.000000	NaN	NaN	NaN	2020-07-28 00:00:00	NaN	NaN	NaN	13241.224655	202.000000	NaN	2020-08-1 00:00:00
50%	NaN	52.000000	NaN	NaN	NaN	2021-11-01 00:00:00	NaN	NaN	NaN	25538.069380	302.000000	NaN	2021-11-1 00:00:00
75%	NaN	68.000000	NaN	NaN	NaN	2023-02-03 00:00:00	NaN	NaN	NaN	37820.508432	401.000000	NaN	2023-02-1 00:00:00
max	NaN	89.000000	NaN	NaN	NaN	2024-05-07 00:00:00	NaN	NaN	NaN	52764.276740	500.000000	NaN	2024-06-0 00:00:00
std	NaN	19.602454	NaN	NaN	NaN	NaN	NaN	NaN	NaN	14211.454431	115.243069	NaN	NaN

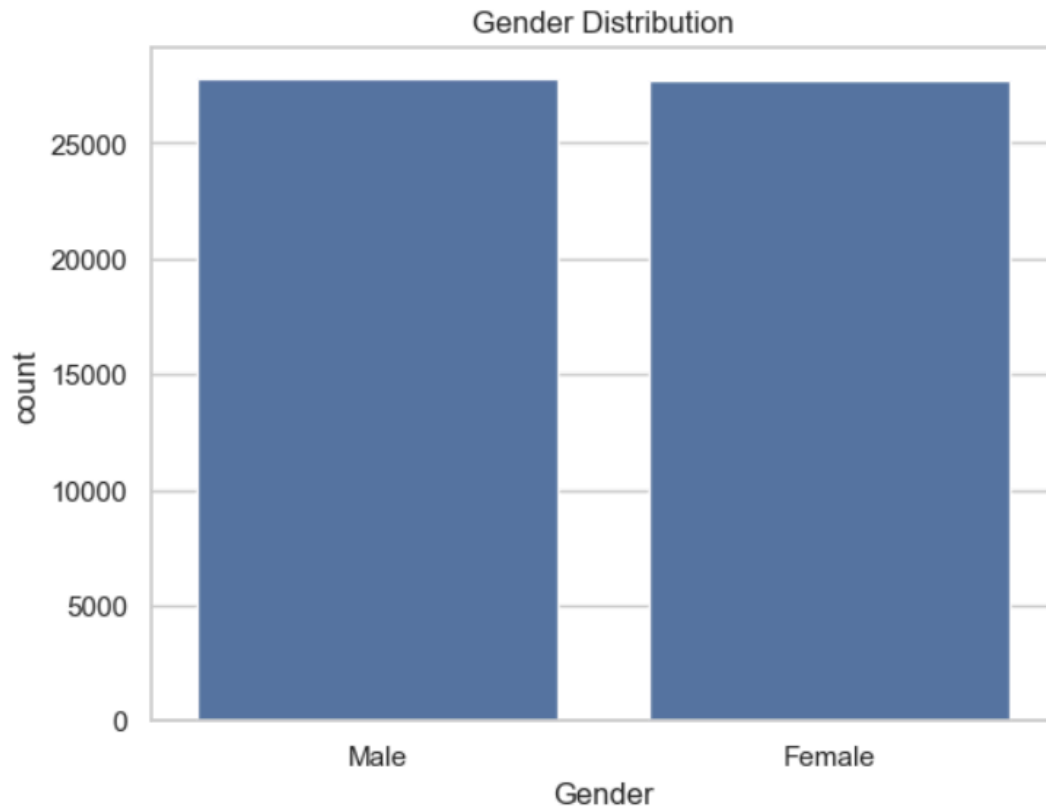
- Found the missing values in the dataset using isnull()

```
Name          0
Age           0
Gender        0
Blood Type    0
Medical Condition  0
Date of Admission  0
Doctor        0
Hospital      0
Insurance Provider  0
Billing Amount  0
Room Number   0
Admission Type  0
Discharge Date  0
Medication    0
Test Results   0
dtype: int64
```

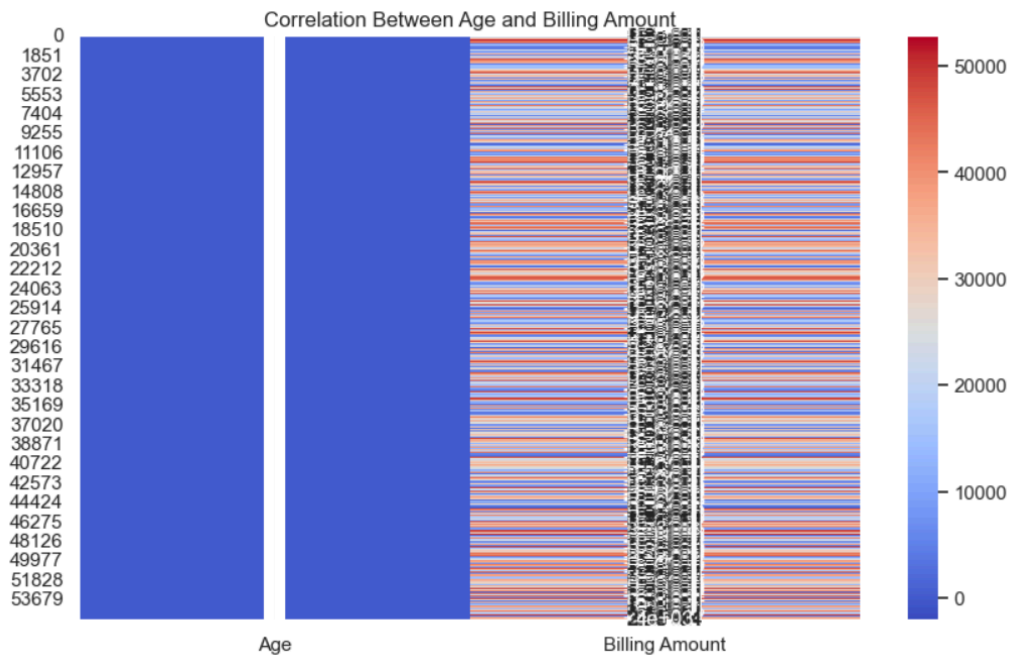
- Distribution analysis of age, gender and billing amount using seaborn library's histplot and countplot.

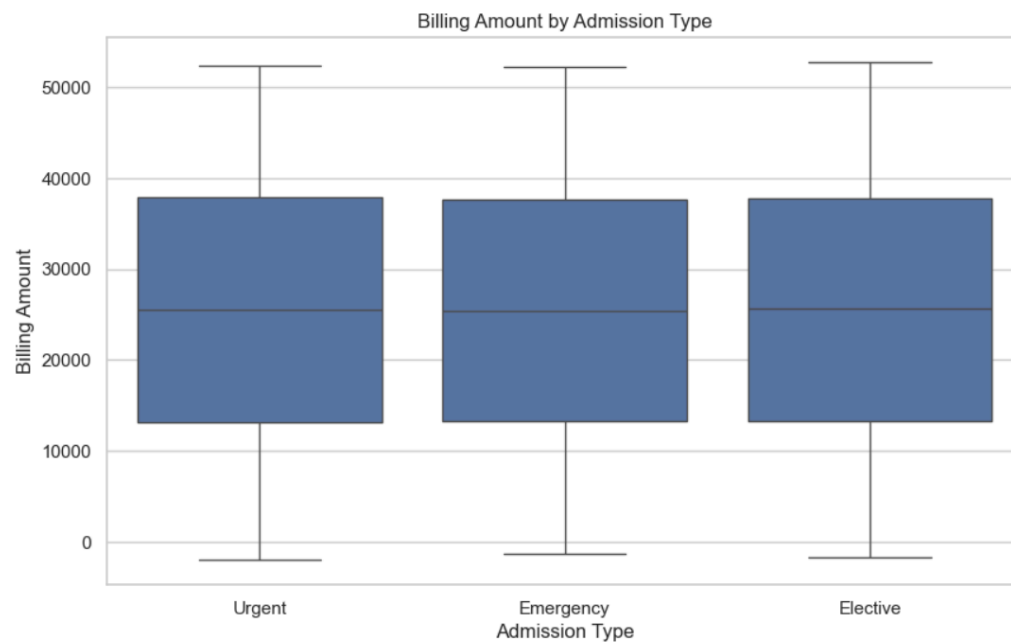
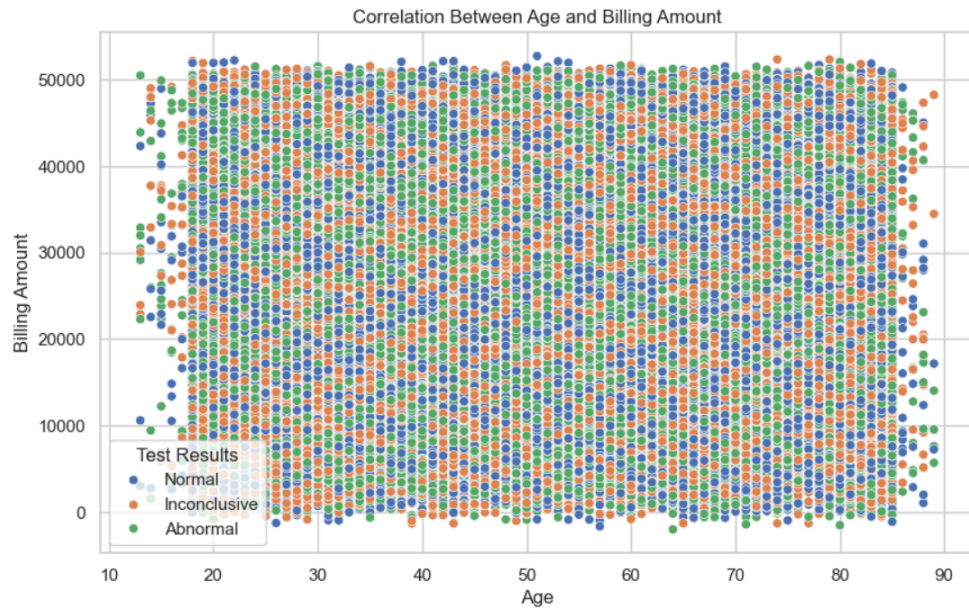






- Finding correlations between variables with heatmap and scatterplot





7. Data Preprocessing

- Handling of the missing values with `fillna()`, `median()`, & `mode()` functions
- Encoding Categorical variables and ensuring consistency in data formatting.

```

# Fill missing numerical values with median
healthcare['Billing Amount'].fillna(healthcare['Billing Amount'].median(), inplace = True)

# Fill missing categorical values with mode
categorical_col = ['Gender', 'Blood Type', 'Medical Condition', 'Doctor', 'Hospital', 'Insurance Provider', 'Admission Type', 'Medication', 'Test Results']
for category in categorical_col:
    healthcare[category].fillna(healthcare[category].mode()[0], inplace = True)

# Handle missing dates
# Convert from object to datetime
healthcare['Date of Admission'] = pd.to_datetime(healthcare['Date of Admission'])
healthcare['Discharge Date'] = pd.to_datetime(healthcare['Discharge Date'], dayfirst = True)

healthcare.isnull().sum()

```

```

Name                0
Age                 0
Gender              0
Blood Type          0
Medical Condition    0
Date of Admission    0
Doctor              0
Hospital            0
Insurance Provider   0
Billing Amount       0
Room Number         0
Admission Type       0
Discharge Date       0
Medication           0
Test Results         0
dtype: int64

```

Feature Engineering

```

5]: # Length of stay in days
healthcare['Length of Stay'] = (healthcare['Discharge Date'] - healthcare['Date of Admission']).dt.days

# Room utilization flag
healthcare['Room Occupied'] = healthcare['Room Number'].notnull().astype(int)

```

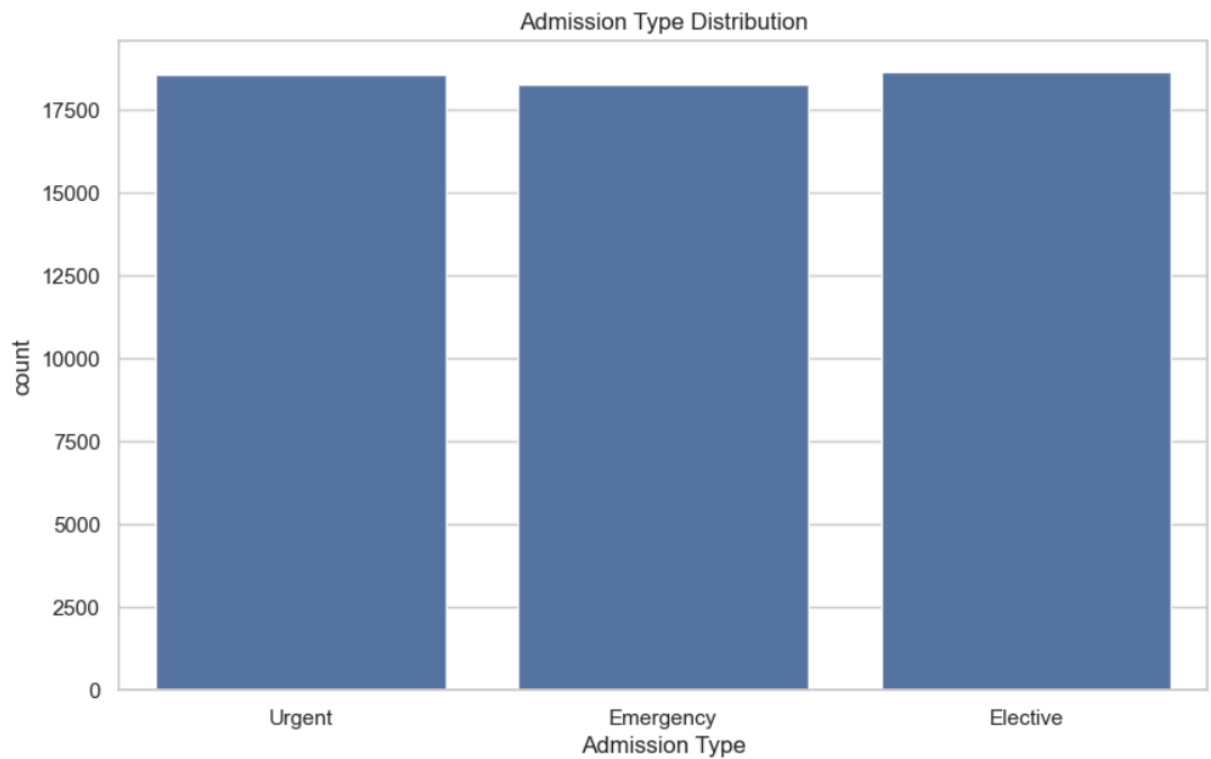
Categorical Encoding

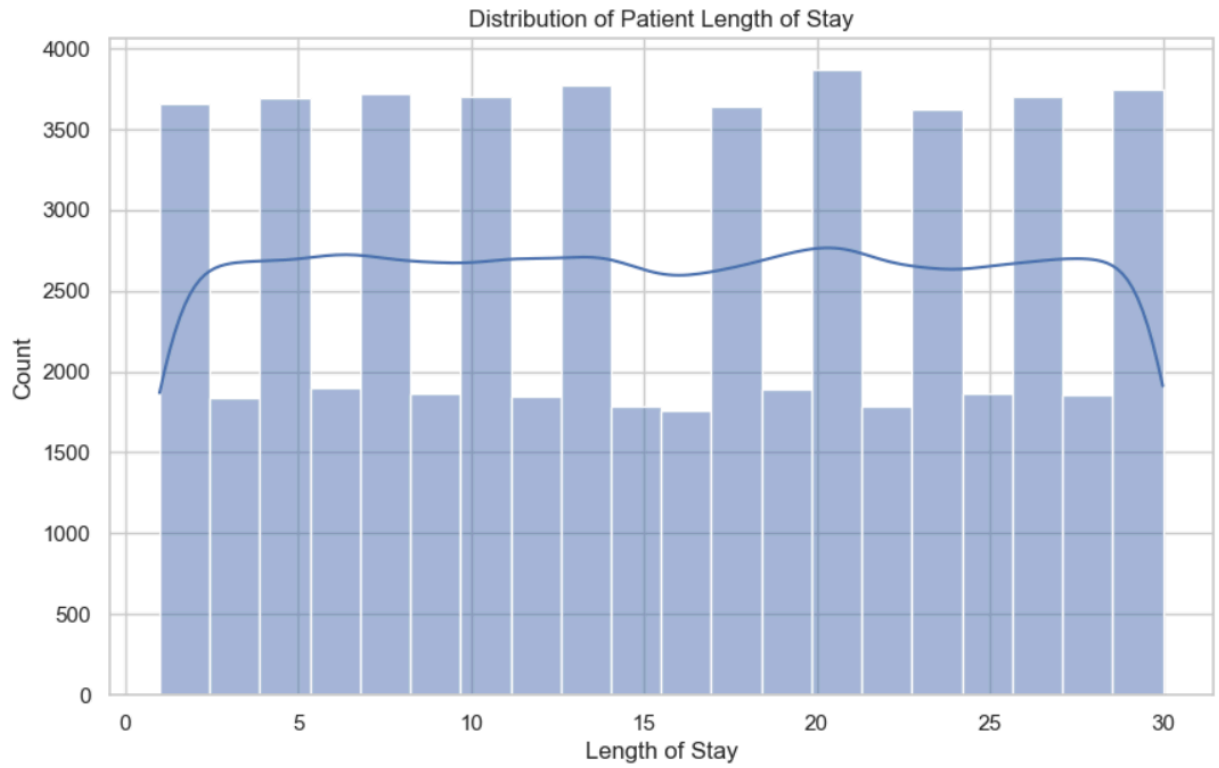
```
# Convert categorical variables to category dtype for analysis/ml

for category in categorical_col:
    healthcare[category] = healthcare[category].astype('category')
```

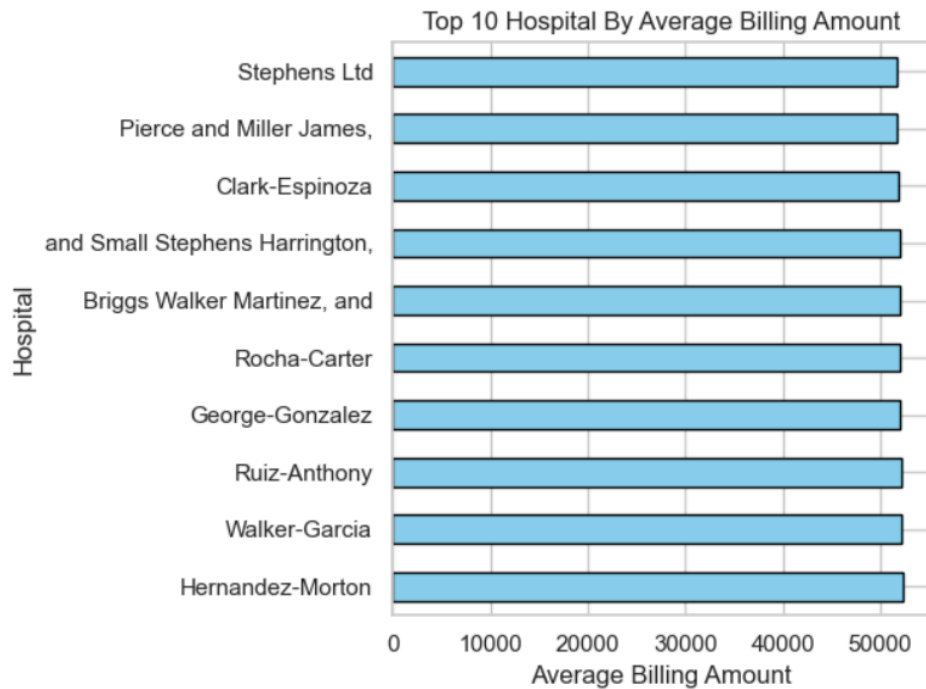
8. Trend Analysis

- Distribution of patients length of stay, admission type.





- Finding average length of stay, average of top 10 hospitals with mean function



```

: # Room utilisation rate
room_utilization_rate = healthcare['Room Occupied'].mean()
print('Room Utilization Rate:', round(room_utilization_rate * 100, 2), '%')

Room Utilization Rate: 100.0 %

```

9. Key Metrics

```

: # Average patient billing amounts
avg_billing = healthcare['Billing Amount'].mean()
print('Average Patient Billing Amount:', round(avg_billing, 2))

# Length of stay analysis
length_of_stay_stats = healthcare['Length of Stay'].describe()
print(length_of_stay_stats)

# Admission type distribution
admission_counts = healthcare['Admission Type'].value_counts()
print(admission_counts)

```

Average Patient Billing Amount: 25539.32

count 55500.000000

mean 15.509009

std 8.659600

min 1.000000

25% 8.000000

50% 15.000000

75% 23.000000

max 30.000000

Name: Length of Stay, dtype: float64

Admission Type

Elective 18655

Urgent 18576

Emergency 18269

Name: count, dtype: int64

10. Business Recommendation.

1. Optimize Room Utilisation: If room utilisation is below optimal (<70%), implement scheduling strategies to reduce empty room days.

2. **Manage Length of Stay:** Identify conditions with longer-than-average stays and review care protocols to reduce unnecessary hospital days.
3. **Control Billing Costs:** Analyze high-billing patients and treatment plans to identify cost-saving opportunities without compromising care.
4. **Admission Type Management:** Track urgent vs. non-urgent admissions to improve resource allocation and emergency preparedness.
5. **Data Standardization:** Maintain consistent data entry for patient demographics, medications, and test results to enable faster reporting and predictive analytics.

11. Executive Summary

This report presents an in-depth analysis of hospital patient data to uncover operational trends, cost drivers, and resource utilization patterns. Using exploratory data analysis (EDA), data preprocessing, and trend analysis techniques, key insights were derived to support decision-making aimed at improving hospital efficiency, cost control, and patient care quality.

The analysis focuses on:

- Patient demographics
- Length of hospital stay
- Admission types
- Billing patterns
- Room utilization

12. Key Findings & Insights

Patient Demographics

Patient age distribution shows a wide range, with most patients falling in the adult and senior categories.

Gender distribution is relatively balanced, suggesting no demographic bias in admissions.

Visuals Used: Age histogram, Gender bar chart

Length of Stay Analysis

The average length of stay indicates most patients are discharged within a reasonable timeframe.

However, outliers exist, where certain patients remain hospitalized significantly longer.

Longer stays are often linked to specific medical conditions or urgent admissions.

Visuals Used: Length of stay histogram, Box plot

Business Insight:

Extended hospital stays increase operational costs and reduce room availability.

Billing Amount Trends

Billing amounts vary widely across patients.

Urgent admissions generally result in higher billing amounts than non-urgent cases.

Certain hospitals exhibit consistently higher average billing, indicating potential differences in service pricing or case severity.

Visuals Used: Billing histogram, Billing by admission type box plot, Hospital-wise bar chart

Business Insight:

Billing variability highlights opportunities for cost optimization and pricing standardization.

Admission Type Distribution

The majority of admissions fall into non-urgent categories.

Urgent admissions, while fewer, consume disproportionately higher resources.

Visuals Used: Admission type bar chart

Business Insight:

Better triaging of non-urgent cases could reduce strain on hospital resources.

Room Utilization

Room utilization rate indicates that not all rooms are optimally used.

Periods of underutilization suggest inefficiencies in admission planning and discharge coordination.

Visuals Used: Room utilization metrics and summaries

Business Insight:

Improving room turnover can significantly increase hospital capacity without infrastructure expansion.

13. Key Performance Metrics

Average Billing Amount - Indicates overall treatment cost per patient

Average Length of Stay - Key driver of operational efficiency

Room Utilization Rate - Measures infrastructure efficiency

Urgent vs Non-Urgent Ratio - Reflects emergency resource demand

These metrics directly support capacity planning, staffing, and financial forecasting.

14. Conclusion

This analysis demonstrates how healthcare data can be leveraged to:

Improve operational efficiency

Reduce unnecessary costs

Enhance patient care quality

Support strategic planning

By acting on the insights and recommendations provided, hospitals can achieve better resource utilization, financial sustainability, and improved patient outcomes.